



ADITYA UNIVERSITY

SCHOOL OF COMPUTING

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Cornerstone project work entitled “**Railway Reservation Management System**” is submitted by

J.Dileep Kumar	(25B11CS342)
Askhay Putta	(25B11CS022)
A.Sasank	(25B11CS027)
S.Hemanth	(25B11CS844)

in partial fulfillment of the requirements for the award of the B.Tech. degree in Computer Science and Engineering for the academic year 2026-2027.

Project Guide

Mrs. K Vydehi, M.Tech.,(Ph.D),
Assistant Professor,
Department of CSE

Head of the Department

Dr. Phani Sridhar Addepalli, Ph.D,
Assistant Professor,
Department of CSE

DECLARATION

We hereby declare that the Cornerstone project entitled “**University Academic Management Database Application**” is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. Degree in Computer Science and Engineering**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

Place: Surampalem

Date:

By

J DILEEP KUMAR	(25B11CS342)
AKSHAY PUTTA	(25B11CS022)
A SASANK	(25B11CS027)
S HEMANTH	(25B11CS844)

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Name of the Students

J DILEEP KUMAR	(25B11CS342)
AKSHAY PUTTA	(25B11CS022)
A SASANK	(25B11CS027)
S HEMANTH	(25B11CS844)

Date:

TABLE OF CONTENTS

CONTENTS	PAGE NO.S
1. Cover Page	i
2. Certificate	ii
3. Acknowledgement	iii
4. Table of Contents	1
5. Abstract	2
6. Introduction	6
7. System Analysis	7
8. Database Design	9
9. ER Diagram	11
10. Relational Schema	12
11. Table Design	13
12. Normalization	16
13. SQL Implementation	19
14. SQL Queries & Screenshots	22
15. Conclusion & Future Enhancements	29
16. References	29

ABSTRACT

The Railway Reservation Management System is a computer-based application designed to automate the booking, management, and cancellation of train tickets, replacing the traditional manual process that is often slow, tedious, and prone to human error, especially as the volume of passengers continues to grow across the rail network. The system is built around a centralized database that stores detailed information on trains, routes, schedules, station lists, fares, and registered passengers, and it allows a passenger to search for available trains between specified source and destination stations, view real-time seat availability across different classes such as sleeper, AC, and general, reserve a seat by entering personal and journey details, generate a printable ticket carrying a unique Passenger Name Record (PNR) for tracking, and cancel an existing booking whenever necessary. Every reservation or cancellation instantly updates the seat inventory in the database, which prevents problems such as double booking or overselling of seats, while a waiting list mechanism automatically promotes and confirms pending tickets as confirmed seats are freed up through cancellations. Fares are computed automatically based on factors such as distance travelled and class of travel, removing the need for manual fare calculation and reducing the chance of billing errors. Beyond the passenger-facing functions, the system includes a dedicated administrative module that grants a system administrator the privilege to add or remove trains, update schedules, routes and fares, monitor and search booking records, and generate useful reports on passenger traffic, occupancy, and revenue over a given period, while ordinary users remain restricted to booking, viewing, and cancelling only their own tickets, ensuring proper data security and separation of roles. Altogether, the system was developed following a structured approach involving requirement analysis, system design, and implementation, using a relational database for reliable data storage and a simple, user-friendly interface for interaction, and the finished project offers a reliable, efficient, and easy-to-use platform that significantly reduces the time spent booking tickets, minimizes human error, eliminates seat duplication, and equips railway authorities with an effective tool for reservation management and reporting, with clear potential for further enhancement through features such as online payment integration, SMS/email notifications, and mobile application accessibility.

LIST OF FIGURES

Si no	Figures	Page no.s
1	ER Diagram of Railway Reservation Management System (Chen Notation)	11
2	Unnormalized Form (UNF) Data	16
3	First Normal Form (1NF) Data	16
4	Second Normal Form (2NF) - Reservation Relation	17
5	Second Normal Form (2NF) - Passenger Phone Relation	17
6	Third Normal Form (3NF) - Reservation Relation	18
7	Third Normal Form (3NF) - Train Relation	18
8	Third Normal Form (3NF) - Passenger Phone Relation	18
9	Output of Q1: Display all passengers	22
10	Output of Q2: Display passenger first name, last name, and gender	23
11	Output of Q3: Display female passengers	23
12	Output of Q4: Trains with more than 400 total seats	24
13	Output of Q5: Available seats per train	24
14	Output of Q6: Average fare across all reservations	24
15	Output of Q7: Total amount collected	25
16	Output of Q8: Number of reservations per class	25
17	Output of Q9: Payment modes collecting more than 3000	26
18	Output of Q10: Reservations sorted by fare (highest first)	26
19	Output of Q11: Passengers whose first name starts with A	27
20	Output of Q12: Highest fare charged	27
21	Output of Q13: Total number of passengers	27
22	Output of Q14: Number of payments per payment mode	28
23	Output of Q15: Reservations with fare above 1000	28

LIST OF TABLES

Si no	Table	Page nos
1	Functional Requirements	7
2	Requirements Specification	8
3	Major Database Entities	9
4	Relationships Between Entities	9
5	Relational Schema	12
6	Passenger Table Structure	13
7	Passenger_Phone Table Structure	13
8	Train Table Structure	13
9	Station Table Structure	13
10	Train_Station Table Structure	13
11	Reservation Table Structure	14
12	Ticket Table Structure	14
13	Payment Table Structure	14
14	Keys and Constraints	15
15	Reasons for Separation in Third Normal Form (3NF)	18

ACRONYMS

- DBMS – Database Management System
- ER – Entity Relationship
- SQL – Structured Query Language
- DDL – Data Definition Language
- DML – Data Manipulation Language
- DQL – Data Query Language
- PK – Primary Key
- FK – Foreign Key
- 1NF – First Normal Form
- 2NF – Second Normal Form
- 3NF – Third Normal Form
- PNR – Passenger Name Record

1. INTRODUCTION

1.1 Brief Information about the Project

A Railway Reservation Management System is a database application that manages railway passenger and reservation information. The project provides a centralized relational database for passengers, trains, stations, reservations, tickets, and payments. It is intended to replace fragmented manual records and disconnected files with a structured database that supports consistent storage and efficient retrieval.

1.2 Motivation and Contribution of Project

The project is motivated by common problems in railway reservation management such as duplicate passenger records, difficulty checking seat availability, time-consuming searches, inconsistent reservation information, and difficulty generating consolidated reports. A centralized database provides a systematic approach to storing related railway data and enforcing relationships through keys and constraints.

- Centralizing railway reservation records in one relational database.
- Reducing duplicate and inconsistent passenger and reservation information.
- Maintaining referential integrity using primary and foreign keys.
- Supporting faster passenger, train, station, and reservation queries.
- Providing a foundation for ticket and payment tracking and reservation reporting.

1.3 Objectives of the Project

- Develop a centralized railway reservation database.
- Manage passenger records, including multiple contact numbers.
- Maintain train and station information.
- Process passenger reservations and store journey, class, fare, seat, and coach details.
- Store ticket (PNR) information linked to each reservation.
- Record payment details linked to each reservation.
- Reduce duplicate data and update anomalies.
- Maintain referential integrity.
- Support efficient SQL queries and railway reports.

1.4 Scope of the Project

The scope covers database-oriented management of railway reservation information. The core scope includes passenger, passenger phone, train, station, train–station route, reservation, ticket, and payment data. The database can be used as the backend for a reservation application and can be extended later with cancellation/refund workflows, wait-list management, notifications, analytics dashboards, and web/mobile interfaces.

2. SYSTEM ANALYSIS

2.1 Existing System

- Manual or semi-manual reservation record maintenance.
- Multiple spreadsheets or disconnected files.
- Duplicate passenger and reservation records.
- Difficulty checking current seat availability.
- Time-consuming searches for train and passenger details.
- Difficulty generating consolidated reservation and payment reports.
- Higher possibility of data-entry and booking errors.

2.2 Proposed System

The proposed system uses a centralized relational database. It organizes railway reservation information into related tables and uses primary keys, foreign keys, uniqueness rules, and appropriate constraints. The system supports passenger, train, station, reservation, ticket, and payment management, with derived attributes such as age and available seats computed through views instead of stored redundantly.

- Centralized database
- Passenger management
- Train management
- Station management
- Reservation processing
- Ticket (PNR) management
- Payment tracking
- SQL-based reporting
- Referential integrity

2.3 Functional Requirements

Table 1: Functional Requirements

Module	Functional Requirement
Passenger Management	Add, update, search, and display passenger details, including multiple phone numbers.
Train Management	Store and maintain train identity, name, type, and seat capacity.
Station Management	Store station name, city, and state details, and the stations served by each train.
Reservation Management	Book seats for passengers and maintain journey, class, fare, seat, and coach details.
Ticket Management	Generate and store PNR and booking date linked to a reservation.
Payment Management	Store payment amount, date, and mode associated with a reservation.
Search and Reporting	Retrieve railway information using SQL queries.

2.4 Non-Functional Requirements

- Performance – queries should return results efficiently for normal reservation workloads.
- Integrity – invalid references and duplicate key values should be prevented.
- Security – database privileges should be controlled for authorized users.
- Usability – forms and reports should be understandable to reservation users.
- Reliability – transactions should preserve a consistent database state.
- Scalability – the design should support growth in passengers, trains, and reservations.

2.5 Requirements Specification

Table 2: Requirements Specification

Category	Suggested Requirement
Hardware	Intel i3 or above; 4 GB RAM minimum; adequate storage; standard display and keyboard/mouse.
Operating System	Windows 10/11 or an equivalent supported platform.
Database	Oracle Database or another relational DBMS used by the implementation.

3. DATABASE DESIGN

3.1 Major Entities

Table 3: Major Database Entities

Entity	Description
Passenger	Stores passenger personal information (name, gender, date of birth). Age is a derived attribute, not stored.
Passenger_Phone	Stores one or more contact phone numbers per passenger (multi-valued attribute of Passenger).
Train	Stores train identity, type, and seat capacity information. Available seats is a derived attribute, not stored.
Station	Stores station name, city, and state information.
Train_Station	Stores the stations served by each train (resolves the many-to-many relationship between Train and Station).
Reservation	Stores journey, class, fare, seat, and coach details for a booking made by a passenger.
Ticket	Stores the PNR and booking date associated with a reservation.
Payment	Stores payment amount, date, and mode associated with a reservation.

3.2 ER Model

The ER model represents the Railway Reservation Management System using Chen-style entity-relationship notation. The major entities shown in the diagram are PASSENGERS, RESERVATION, TRAIN, TICKET, PAYMENT, and STATION. The diagram uses rectangles for entities, diamonds for relationships, ovals for attributes, underlined attributes for key attributes, a dashed oval for the derived AGE and AVAILABLE_SEATS attributes, a double oval for the multi-valued PHONE_NO attribute, and composite attributes NAME (FIRST_NAME, LAST_NAME), SEAT_ID (COACH_NO, SEAT_NUMBER) and SEATS (TOTAL_SEATS, BOOKED_SEATS, AVAILABLE_SEATS). The principal relationships are PASSENGER-RESERVATION (MAKES), TRAIN-RESERVATION (BOOKS), RESERVATION-TICKET (INCLUDE), RESERVATION-PAYMENT (GENERATES), and TRAIN-STATION (ARRIVES). The many-to-many ARRIVES relationship is implemented in the relational schema as a separate Train_Station table.

3.3 Relationships

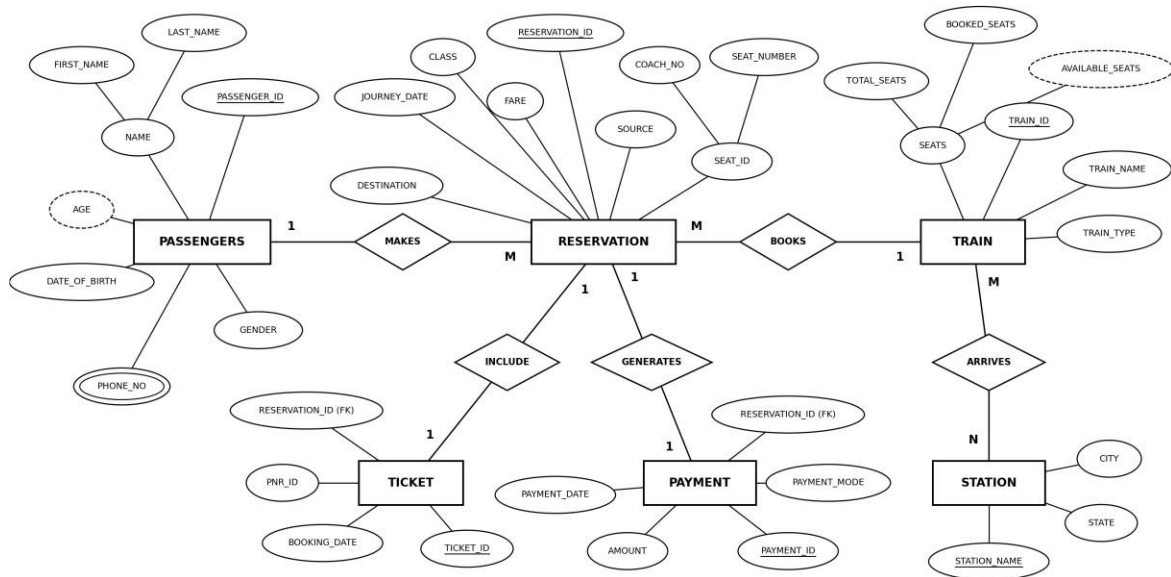
Table 4: Relationships Between Entities

Relationship	Description
Passenger → Reservation	One passenger can make multiple reservations.
Train → Reservation	One train can have multiple reservations.
Reservation → Ticket	A reservation includes a ticket.
Reservation → Payment	A reservation generates a payment record.

Relationship	Description
Train ↔ Station	A train can arrive at multiple stations, and a station can be served By multiple trains.This many-to-many relationship is implemented through the Train_Station table.

4. ER DIAGRAM

Figure 1: ER Diagram of Railway Reservation Management System (Chen Notation)



The ER diagram summarizes the entity structure and cardinalities used by the proposed railway reservation database. The entities and attributes shown in the diagram — Passenger (with Passenger_ID, Name, Gender, Date_of_Birth, Age, Phone_No), Reservation, Train, Ticket, Payment, and Station — are directly reflected in the relational schema, table design, and SQL implementation in the following sections.

5. RELATIONAL SCHEMA

Table 5: Relational Schema

Relation	Relational Schema
PASSENGER	Passenger_ID (PK), First_Name, Last_Name, Gender, Date_of_Birth
PASSENGER_PHONE	Passenger_ID (PK, FK), Phone_No (PK)
TRAIN	Train_ID (PK), Train_Name, Train_Type, Total_Seats, Booked_Seats
STATION	Station_Name (PK), State, City
TRAIN_STATION	Train_ID (PK, FK), Station_Name (PK, FK)
RESERVATION	Reservation_ID(PK),Passenger_ID (FK), Train_ID (FK), Journey_Date, Source, Destination, Class, Fare, Seat_Number, Coach_No
TICKET	Ticket_ID (PK), PNR_ID, Booking_Date, Reservation_ID (FK)
PAYMENT	Payment_ID(PK),Amount,Payment_Date,Payment_Mode, eservation_ID (FK)

The relational schema separates passenger and railway master data from reservation, ticket, and payment transaction data. Passenger_Phone references Passenger; Train_Station references Train and Station; Reservation references Passenger and Train; Ticket references Reservation; and Payment references Reservation.

6. TABLE DESIGN

6.1 Passenger Table

Table 6: Passenger Table Structure

Attribute	Data Type	Constraint	Description
Passenger_ID	VARCHAR2(10)	PK	Unique passenger identifier
First_Name	VARCHAR2(30)	NOT NULL	Passenger first name
Last_Name	VARCHAR2(30)	NOT NULL	Passenger last name
Gender	CHAR(1)	CHECK (M, F)	Passenger gender
Date_of_Birth	DATE	NOT NULL	Used to derive age via view

6.2 Passenger_Phone Table

Table 7: Passenger_Phone Table Structure

Attribute	Data Type	Constraint	Description
Passenger_ID	VARCHAR2(10)	PK, FK → Passenger	References Passenger
Phone_No	VARCHAR2(15)	PK	One of possibly several contact numbers

6.3 Train Table

Table 8: Train Table Structure

Attribute	Data Type	Constraint	Description
Train_ID	VARCHAR2(10)	PK	Unique train identifier
Train_Name	VARCHAR2(50)	NOT NULL	Train name
Train_Type	VARCHAR2(20)	—	Superfast, Express, etc.
Total_Seats	NUMBER	CHECK (> 0)	Total seat capacity
Booked_Seats	NUMBER	DEFAULT 0, CHECK (0 to Total_Seats)	Seats currently booked; used to derive available seats via view

6.4 Station Table

Table 9: Station Table Structure

Attribute	Data Type	Constraint	Description
Station_Name	VARCHAR2(50)	PK	Station name (natural key)
State	VARCHAR2(50)	NOT NULL	State
City	VARCHAR2(50)	NOT NULL	City

6.5 Train_Station Table

Table 10: Train_Station Table Structure

Attribute	Data Type	Constraint	Description
Train_ID	VARCHAR2(10)	PK, FK → Train	References Train
Station_Name	VARCHAR2(50)	PK, FK → Station	References Station

6.6 Reservation Table

Table 11: Reservation Table Structure

Attribute	Data Type	Constraint	Description
Reservation_ID	VARCHAR2(10)	PK	Unique reservation identifier
Passenger_ID	VARCHAR2(10)	FK → Passenger, NOT NULL	References Passenger
Train_ID	VARCHAR2(10)	FK → Train, NOT NULL	References Train
Journey_Date	DATE	NOT NULL	Date of journey
Source	VARCHAR2(50)	NOT NULL	Source station
Destination	VARCHAR2(50)	NOT NULL	Destination station
Class	VARCHAR2(20)	CHECK (AC1, AC2, AC3, Sleeper)	AC1, AC2, AC3, Sleeper, etc.
Fare	NUMBER(10,2)	CHECK (> 0)	Ticket fare
Seat_Number	VARCHAR2(10)	—	Assigned seat number
Coach_No	VARCHAR2(10)	—	Assigned coach number

6.7 Ticket Table

Table 12: Ticket Table Structure

Attribute	Data Type	Constraint	Description
Ticket_ID	VARCHAR2(10)	PK	Unique ticket identifier
PNR_ID	VARCHAR2(20)	UNIQUE, NOT NULL	Passenger Name Record
Booking_Date	DATE	—	Date of booking
Reservation_ID	VARCHAR2(10)	FK → Reservation, UNIQUE, NOT NULL	References Reservation

6.8 Payment Table

Table 13: Payment Table Structure

Attribute	Data Type	Constraint	Description
Payment_ID	VARCHAR2(10)	PK	Unique payment identifier
Amount	NUMBER(10,2)	CHECK (> 0)	Payment amount
Payment_Date	DATE	—	Date of payment
Payment_Mode	VARCHAR2(20)	CHECK (UPI, Card, NetBanking)	UPI, Card, NetBanking
Reservation_ID	VARCHAR2(10)	FK → Reservation, UNIQUE, NOT NULL	References Reservation

6.9 Keys and Constraints

Table 14: Keys and Constraints

Constraint Type	Applied To / Purpose
Primary Key	Passenger_ID; Passenger_ID+Phone_No; Train_ID; Station_Name; Train_ID + Station_Name; Reservation_ID; Ticket_ID; Payment_ID
Foreign Key	Passenger_Phone.Passenger_ID; Train_Station.Train_ID; Train_Station.Station_Name; Reservation.Passenger_ID; Reservation.Train_ID; Ticket.Reservation_ID; Payment.Reservation_ID
NOT NULL	Passenger names and date of birth; train name; station state and city; passenger, train, journey date, source and destination in Reservation; PNR and reservation in Ticket; reservation in Payment
UNIQUE	Ticket.PNR_ID; Ticket.Reservation_ID; Payment.Reservation_ID (one ticket and one payment per reservation)
CHECK	Gender (M, F); Total_Seats > 0; Booked_Seats between 0 and Total_Seats; Class values; Fare > 0; Amount > 0; Payment_Mode values
DEFAULT	Train.Booked_Seats defaults to 0

7. NORMALIZATION

7.1 Unnormalized Form (UNF)

A single railway reservation record (railway_unf) stores Passenger_ID, name, age, gender, DOB, along with comma-separated repeating lists for phone numbers, reservation IDs, journey dates, sources, destinations, classes, fares, seat numbers, coach numbers, train details, seat-count details, PNR IDs, booking dates, ticket IDs, payment IDs, amounts, payment details, and station details — all in one row per passenger. This repeating-group structure violates 1NF.

Figure 2: Unnormalized Form (UNF) Data

Results

Explain

Describe

Saved SQL

History

RESERVATION_ID	PASSENGER_ID	PHONE_NO	TRAIN_ID	TRAIN_NAME	FARE
R101	P1	9876543210, 9123456780	T12	Godavari Exp	450
R102	P1	9876543210, 9123456780	T15	Rayalaseema Exp	300
R103	P2	9000011111	T12	Godavari Exp	470
R104	P3	9988776655, 9012345678	T18	Venkatadri Exp	520
R105	P4	9700012345	T15	Rayalaseema Exp	320
R106	P2	9000011111	T18	Venkatadri Exp	540

6 rows returned in 0.01 seconds

CSV Export

7.2 First Normal Form (1NF)

Repeating groups are removed by splitting each passenger's record into one row per reservation (railway_1nf, composite key: Passenger_ID + Reservation_ID). The multi-valued Phone_No attribute is pulled into its own table, passenger_phones_1nf (Passenger_ID + Phone_No). All remaining attributes hold atomic values, but train, station, and payment details are still repeated across rows for the same passenger or train.

Figure 3: First Normal Form (1NF) Data

Results

Explain

Describe

Saved SQL

History

RESERVATION_ID	PASSENGER_ID	PHONE_NO	TRAIN_ID	TRAIN_NAME	FARE
R101	P1	9876543210	T12	Godavari Exp	450
R101	P1	9123456780	T12	Godavari Exp	450
R102	P1	9876543210	T15	Rayalaseema Exp	300
R102	P1	9123456780	T15	Rayalaseema Exp	300
R103	P2	9000011111	T12	Godavari Exp	470
R104	P3	9988776655	T18	Venkatadri Exp	520
R104	P3	9012345678	T18	Venkatadri Exp	520
R105	P4	9700012345	T15	Rayalaseema Exp	320
R106	P2	9000011111	T18	Venkatadri Exp	540

9 rows returned in 0.00 seconds

CSV Export

7.3 Second Normal Form (2NF)

Partial dependencies on the composite key are removed. Attributes that depend only on Passenger_ID (name, gender, DOB) are moved into passengers_2nf; phone numbers stay in passenger_phones_2nf referencing it by foreign key. The remaining reservation-specific and repeated train/station/payment attributes stay together in reservations_2nf, keyed by Reservation_ID — but this still carries transitive dependencies, since train and station details depend on Train_ID/Station_Name rather than directly on Reservation_ID.

Figure 4: Second Normal Form (2NF) - Reservation Relation

RESERVATION_ID	PASSENGER_ID	TRAIN_ID	TRAIN_NAME	FARE
R101	P1	T12	Godavari Exp	450
R102	P1	T15	Rayalaseema Exp	300
R103	P2	T12	Godavari Exp	470
R104	P3	T18	Venkatadri Exp	520
R105	P4	T15	Rayalaseema Exp	320
R106	P2	T18	Venkatadri Exp	540

Figure 5: Second Normal Form (2NF) - Passenger Phone Relation

PASSENGER_ID	PHONE_NO
P1	9876543210
P1	9123456780
P2	9000011111
P3	9988776655
P3	9012345678
P4	9700012345

1.1 Third Normal Form (3NF)

Transitive dependencies are removed by extracting train and station details into their own relations:

- passengers_3nf – passenger attributes only; Age is removed and replaced by the derived view v_passenger_age (computed from Date_of_Birth using MONTHS_BETWEEN).
- passenger_phones_3nf – unchanged from 2NF, foreign key to Passenger.
- trains_3nf – train attributes only; Available_Seats is removed and replaced by the derived view v_train_seats (computed as Total_Seats – Booked_Seats).
- stations_3nf – station attributes only, keyed by Station_Name.
- train_station_3nf – stations served by each train; composite key of Train_ID and Station_Name, with foreign keys to Train and Station.

- reservations_3nf – reservation-specific attributes, with foreign keys to Passenger and Train.
- tickets_3nf – PNR and booking date, separated out and linked to Reservation by foreign key.
- payments_3nf – payment attributes, linked to Reservation by foreign key.

Table 15: Reasons for Separation in Third Normal Form (3NF)

Relation	Reason for Separation
Passenger	Passenger-specific attributes only; age handled as a derived view.
Passenger_Phone	Multi-valued phone numbers separated to satisfy 1NF/2NF.
Train	Train-specific attributes only; available seats handled as a derived view.
Station	Station-specific attributes, removed from repeating in every reservation.
Train_Station	Resolves the many-to-many relationship between Train and Station.
Reservation	Booking-specific attributes and foreign keys to Passenger and Train.
Ticket	PNR/booking details separated from Reservation to avoid repetition.
Payment	Payment details separated from Reservation, one payment per reservation.

Figure 6: Third Normal Form (3NF) - Reservation Relation

Figure 7: Third Normal Form (3NF) - Train Relation

RESERVATION_ID	PASSENGER_ID	TRAIN_ID	FARE	TRAIN_ID	TRAIN_NAME
R101	P1	T12	450	T12	Godavari Exp
R102	P1	T15	300	T15	Rayalaseema Exp
R103	P2	T12	470	T18	Venkatadri Exp
R104	P3	T18	520		
R105	P4	T15	320		
R106	P2	T18	540		

Figure 8: Third Normal Form (3NF) - Passenger Phone Relation

PASSENGER_ID	PHONE_NO
P1	9876543210
P1	9123456780
P2	9000011111
P3	9988776655
P3	9012345678
P4	9700012345

2. SQL IMPLEMENTATION

2.1 DDL – Data Definition Language

```
CREATE TABLE Passenger (  
    Passenger_ID VARCHAR2(10) PRIMARY KEY,  
    First_Name VARCHAR2(30) NOT NULL,  
    Last_Name VARCHAR2(30) NOT NULL,  
    Gender CHAR(1),  
    Date_of_Birth DATE NOT NULL,  
    CONSTRAINT chk_gender CHECK (Gender IN ('M','F'))  
);  
  
CREATE TABLE Passenger_Phone (  
    Passenger_ID VARCHAR2(10),  
    Phone_No VARCHAR2(15),  
    CONSTRAINT pk_phones PRIMARY KEY (Passenger_ID, Phone_No),  
    CONSTRAINT fk_phones FOREIGN KEY (Passenger_ID) REFERENCES Passenger(Passenger_ID)  
);  
  
CREATE TABLE Train (  
    Train_ID VARCHAR2(10) PRIMARY KEY,  
    Train_Name VARCHAR2(50) NOT NULL,  
    Train_Type VARCHAR2(20),  
    Total_Seats NUMBER,  
    Booked_Seats NUMBER DEFAULT 0,  
    CONSTRAINT chk_total_seats CHECK (Total_Seats > 0),  
    CONSTRAINT chk_booked_seats CHECK (Booked_Seats >= 0 AND Booked_Seats <= Total_Seats)  
);  
  
CREATE OR REPLACE VIEW v_train_seats AS  
SELECT Train_ID, Train_Name, Train_Type, Total_Seats, Booked_Seats,  
       (Total_Seats - Booked_Seats) AS Available_Seats  
FROM Train;  
  
CREATE TABLE Station (  
    Station_Name VARCHAR2(50) PRIMARY KEY,  
    State VARCHAR2(50) NOT NULL,  
    City VARCHAR2(50) NOT NULL  
);  
  
CREATE TABLE Train_Station (  
    Train_ID VARCHAR2(10),  
    Station_Name VARCHAR2(50),  
    CONSTRAINT pk_train_station PRIMARY KEY (Train_ID, Station_Name),  
    CONSTRAINT fk_ts_train FOREIGN KEY (Train_ID) REFERENCES Train(Train_ID),  
    CONSTRAINT fk_ts_station FOREIGN KEY (Station_Name) REFERENCES Station(Station_Name)  
);  
  
CREATE TABLE Reservation (  
    Reservation_ID VARCHAR2(10) PRIMARY KEY,  
    Passenger_ID VARCHAR2(10) NOT NULL,  
    Train_ID VARCHAR2(10) NOT NULL,  
    Journey_Date DATE NOT NULL,  
    Source VARCHAR2(50) NOT NULL,  
    Destination VARCHAR2(50) NOT NULL,  
    Class VARCHAR2(20),  
    Fare NUMBER(10,2),  
    Seat_Number VARCHAR2(10),  
    Coach_No VARCHAR2(10),  
    CONSTRAINT chk_class CHECK (Class IN ('AC1','AC2','AC3','Sleeper')),  
    CONSTRAINT chk_fare CHECK (Fare > 0),
```

```

CONSTRAINT fk_res_passenger FOREIGN KEY (Passenger_ID) REFERENCES Passenger(Passenger_ID),
CONSTRAINT fk_res_train FOREIGN KEY (Train_ID) REFERENCES Train(Train_ID)
);

CREATE TABLE Ticket (
    Ticket_ID    VARCHAR2(10) PRIMARY KEY,
    PNR_ID       VARCHAR2(20) NOT NULL UNIQUE,
    Booking_Date DATE,
    Reservation_ID VARCHAR2(10) NOT NULL UNIQUE,
    CONSTRAINT fk_ticket_res FOREIGN KEY (Reservation_ID) REFERENCES Reservation(Reservation_ID)
);

CREATE TABLE Payment (
    Payment_ID    VARCHAR2(10) PRIMARY KEY,
    Amount         NUMBER(10,2),
    Payment_Date   DATE,
    Payment_Mode   VARCHAR2(20),
    Reservation_ID VARCHAR2(10) NOT NULL UNIQUE,
    CONSTRAINT chk_amount CHECK (Amount > 0),
    CONSTRAINT chk_pay_mode CHECK (Payment_Mode IN ('UPI','Card','NetBanking')),
    CONSTRAINT fk_payment_res FOREIGN KEY (Reservation_ID) REFERENCES Reservation(Reservation_ID)
);

CREATE OR REPLACE VIEW v_passenger_age AS
SELECT Passenger_ID, First_Name, Last_Name, Date_of_Birth,
       FLOOR(MONTHS_BETWEEN(SYSDATE, Date_of_Birth) / 12) AS Age
FROM Passenger;

```

2.2 DML – Data Manipulation Language

```

-- Passenger
INSERT INTO Passenger VALUES ('P001','Rahul','Sharma','M',DATE '1996-04-12');
INSERT INTO Passenger VALUES ('P002','Priya','Verma','F',DATE '1991-08-23');

-- Passenger_Phone
INSERT INTO Passenger_Phone VALUES ('P001','9876543210');
INSERT INTO Passenger_Phone VALUES ('P001','9876500001');

-- Train
INSERT INTO Train VALUES ('T001','Rajdhani Express','Superfast',500,420);
INSERT INTO Train VALUES ('T002','Shatabdi Express','Express',300,250);
INSERT INTO Train VALUES ('T003','Durgam Express','Superfast',450,400);

-- Station
INSERT INTO Station VALUES ('New Delhi','Delhi','New Delhi');
INSERT INTO Station VALUES ('Mumbai Central','Maharashtra','Mumbai');
INSERT INTO Station VALUES ('Chennai Central','Tamil Nadu','Chennai');
INSERT INTO Station VALUES ('Bangalore City','Karnataka','Bangalore');
INSERT INTO Station VALUES ('Howrah','West Bengal','Kolkata');

-- Train_Station
INSERT INTO Train_Station VALUES ('T001','New Delhi');
INSERT INTO Train_Station VALUES ('T001','Mumbai Central');
INSERT INTO Train_Station VALUES ('T002','Chennai Central');

-- Reservation (Reservation_ID, Passenger_ID, Train_ID, Journey_Date, Source, Destination, Class, Fare, Seat_Number, Coach_No)

```

```

INSERT INTO Reservation VALUES ('R001','P001','T001',DATE '2026-01-10','New Delhi','Mumbai Central','AC1',4500,'12','A1');
INSERT INTO Reservation VALUES ('R002','P001','T003',DATE '2026-02-14','Howrah','New Delhi','AC2',3200,'25','B2');
INSERT INTO Reservation VALUES ('R003','P002','T002',DATE '2026-01-20','Chennai Central','Bangalore City','AC3',1200,'8','C1');
INSERT INTO Reservation VALUES ('R004','P003','T001',DATE '2026-01-25','New Delhi','Mumbai Central','Sleeper',800,'44','S3');

-- Ticket
INSERT INTO Ticket VALUES ('TCK001','PNR100001',DATE '2026-01-01','R001');
INSERT INTO Ticket VALUES ('TCK002','PNR100002',DATE '2026-02-01','R002');
INSERT INTO Ticket VALUES ('TCK003','PNR100003',DATE '2026-01-05','R003');

-- Payment
INSERT INTO Payment VALUES ('PAY001',4500,DATE '2026-01-01','UPI','R001');
INSERT INTO Payment VALUES ('PAY002',3200,DATE '2026-02-01','Card','R002');
INSERT INTO Payment VALUES ('PAY003',1200,DATE '2026-01-05','NetBanking','R003');
INSERT INTO Payment VALUES ('PAY004',800,DATE '2026-01-10','UPI','R004');

```

2.3 DQL – Data Query Language

```

SELECT * FROM Passenger;

SELECT Passenger_ID, First_Name, Last_Name FROM Passenger WHERE Gender = 'M';

SELECT Train_ID, Train_Name, Train_Type FROM Train ORDER BY Train_Name;

SELECT p.First_Name, p.Last_Name, r.Fare
FROM Passenger p
JOIN Reservation r ON p.Passenger_ID = r.Passenger_ID;

SELECT r.Reservation_ID, t.Train_Name, r.Journey_Date
FROM Reservation r
JOIN Train t ON r.Train_ID = t.Train_ID;

SELECT r.Reservation_ID, s.City, s.State
FROM Reservation r
JOIN Station s ON r.Destination = s.Station_Name;

SELECT tk.PNR_ID, r.Fare
FROM Ticket tk
JOIN Reservation r ON tk.Reservation_ID = r.Reservation_ID;

SELECT pay.Payment_ID, pay.Amount, pay.Payment_Mode
FROM Payment pay
JOIN Reservation r ON pay.Reservation_ID = r.Reservation_ID;

```

3.SQL QUERIES & SCREENSHOTS

The following 17 queries are executed on the tables created in Section 8. Q1 to Q15 are data retrieval queries, Q16 is an UPDATE example and Q17 is a DELETE example.

Q1. Display all passengers

```
SELECT * FROM Passenger;
```

Output:

Figure 9: Output of Q1: Display all passengers

Results	Explain	Describe	Saved SQL	History
PASSENGER_ID	FIRST_NAME	LAST_NAME	GENDER	DATE_OF_BIRTH
P001	Rahul	Sharma	M	12-APR-96
P002	Priya	Verma	F	23-AUG-91
P003	Amit	Singh	M	15-JAN-84
P004	Sneha	Reddy	F	02-NOV-99
P005	Vikram	Rao	M	30-JUN-88
P006	Anjali	Nair	F	19-MAR-95
P007	Karan	Mehta	M	05-SEP-80
P008	Divya	Iyer	F	11-DEC-97
P009	Rohan	Gupta	M	08-JUL-92
P010	Neha	Kapoor	F	27-FEB-01

Q2. Display passenger first name, last name, and gender

```
SELECT first_name, last_name, gender FROM Passenger;
```

Output:

Figure 10: Output of Q2: Display passenger first name, last name, and gender

Results	Explain	Describe	Saved SQL	History
FIRST_NAME	LAST_NAME	GENDER		
Rahul	Sharma	M		
Priya	Verma	F		
Amit	Singh	M		
Sneha	Reddy	F		
Vikram	Rao	M		
Anjali	Nair	F		
Karan	Mehta	M		
Divya	Iyer	F		
Rohan	Gupta	M		
Neha	Kapoor	F		

10 rows returned in 0.00 seconds [CSV Export](#)

Q3. Display female passengers

```
SELECT * FROM Passenger WHERE gender = 'F';
```

Output:

Figure 11: Output of Q3: Display female passengers

Results	Explain	Describe	Saved SQL	History
PASSENGER_ID	FIRST_NAME	LAST_NAME	GENDER	DATE_OF_BIRTH
P002	Priya	Verma	F	23-AUG-91
P004	Sneha	Reddy	F	02-NOV-99
P006	Anjali	Nair	F	19-MAR-95
P008	Divya	Iyer	F	11-DEC-97
P010	Neha	Kapoor	F	27-FEB-01

5 rows returned in 0.00 seconds [CSV Export](#)

Q4. Trains with more than 400 total seats

```
SELECT * FROM Train WHERE total_seats > 400;
```

Output:

Figure 12: Output of Q4: Trains with more than 400 total seats

Results	Explain	Describe	Saved SQL	History
TRAIN_ID	TRAIN_NAME	TRAIN_TYPE	TOTAL_SEATS	BOOKED_SEATS
T001	Rajdhani Express	Superfast	500	420
T003	Duronto Express	Superfast	450	400

2 rows returned in 0.02 seconds [CSV Export](#)

Q5. Available seats per train

SELECT train_name, (total_seats - booked_seats) AS available_seats
FROM Train;

Output:

Figure 13: Output of Q5: Available seats per train

Results Explain Describe Saved SQL History

TRAIN_NAME	AVAILABLE_SEATS
Rajdhani Express	80
Shatabdi Express	50
Duronto Express	50

3 rows returned in 0.01 seconds

CSV Export

Q6. Average fare across all reservations

SELECT AVG(fare) AS average_fare FROM Reservation;

Output:

Figure 14: Output of Q6: Average fare across all reservations

Results

Explain

Describe

Saved SQL

History

AVERAGE_FARE

2310

1 rows returned in 0.00 seconds

CSV Export

Q7. Total amount collected

Figure 17: Output of Q9: Payment modes collecting more than 3000

Results

Explain

Describe

Saved SQL

History

PAYMENT_MODE	TOTAL_PAYMENT
Card	9050
UPI	23550

2 rows returned in 0.01 seconds

[CSV Export](#)

Q10. Reservations sorted by fare (highest first)

```
SELECT * FROM Reservation
```

ORDER BY fare DESC;

Output:

Figure 18: Output of Q10: Reservations sorted by fare (highest first)

Results

Explain

Describe

Saved SQL

History

Reservation_ID	Passenger_ID	Journey_Date	Source	Destination	Class	Fare	Seat_Number	Coach_No	Train_ID	Station_Name
R005	P003	05-MAR-26	Howrah	New Delhi	AC1	4700	3	A1	T003	New Delhi
R013	P009	01-MAR-26	New Delhi	Mumbai Central	AC1	4600	2	A1	T001	Mumbai Central
R001	P001	10-JAN-26	New Delhi	Mumbai Central	AC1	4500	12	A1	T001	Mumbai Central
R011	P008	08-FEB-26	Howrah	New Delhi	AC2	3300	18	B3	T003	New Delhi
R002	P001	14-FEB-26	Howrah	New Delhi	AC2	3200	25	B2	T003	New Delhi
R007	P005	18-JAN-26	New Delhi	Mumbai Central	AC2	3100	15	B1	T001	Mumbai Central
R008	P006	20-FEB-26	Howrah	New Delhi	AC3	2100	22	C2	T003	New Delhi
R015	P010	08-MAR-26	Chennai Central	Bangalore City	AC2	1900	11	B1	T002	Bangalore City
R009	P006	12-MAR-26	Chennai Central	Bangalore City	AC1	1800	5	A1	T002	Bangalore City
R012	P009	22-JAN-26	Chennai Central	Bangalore City	AC3	1250	9	C1	T002	Bangalore City
R003	P002	20-JAN-26	Chennai Central	Bangalore City	AC3	1200	8	C1	T002	Bangalore City
R010	P007	30-JAN-26	New Delhi	Mumbai Central	Sleeper	850	71	S2	T001	Mumbai Central
R004	P003	25-JAN-26	New Delhi	Mumbai Central	Sleeper	800	44	S3	T001	Mumbai Central
R014	P010	25-FEB-26	Howrah	New Delhi	Sleeper	750	55	S1	T003	New Delhi
R006	P004	02-FEB-26	Chennai Central	Bangalore City	Sleeper	600	60	S1	T002	Bangalore City

15 rows returned in 0.00 seconds

[CSV Export](#)

Q11. Passengers whose first name starts with A

```
SELECT * FROM Passenger
```

WHERE first name LIKE 'A%';

Output:

Figure 19: Output of Q11: Passengers whose first name starts with A

Results	Explain	Describe	Saved SQL	History
PASSENGER_ID	FIRST_NAME	LAST_NAME	GENDER	DATE_OF_BIRTH
P003	Amit	Singh	M	15-JAN-84
P006	Anjali	Nair	F	19-MAR-95

2 rows returned in 0.00 seconds [CSV Export](#)

Q12. Highest fare charged

SELECT MAX(fare) AS highest_fare FROM Reservation;

Output:

Figure 20: Output of Q12: Highest fare charged

Results	Explain	Describe	Saved SQL	History
HIGHEST_FARE				
4700				

1 rows returned in 0.02 seconds [CSV Export](#)

Q13. Total number of passengers

SELECT COUNT(*) AS total_passengers FROM Passenger;

Output:

Figure 21: Output of Q13: Total number of passengers

Results	Explain	Describe	Saved SQL	History
TOTAL_PASSENGERS				
10				

1 rows returned in 0.01 seconds [CSV Export](#)

Q14. Number of payments per payment mode

SELECT payment_mode, COUNT(*) AS num_reservations
FROM Payment
GROUP BY payment_mode;

Output:

Figure 22: Output of Q14: Number of payments per payment mode

Results	Explain	Describe	Saved SQL	History
PAYMENT_MODE		NUM_RESERVATIONS		
Card		5		
NetBanking		2		
UPI		8		

3 rows returned in 0.02 seconds [CSV Export](#)

Q15. Reservations with fare above 1000

SELECT * FROM Reservation

WHERE fare > 1000;

Output:

Figure 23: Output of Q15: Reservations with fare above 1000

Results

Explain

Describe

Saved SQL

History

RESERVATION_ID	PASSENGER_ID	JOURNEY_DATE	SOURCE	DESTINATION	CLASS	FARE	SEAT_NUMBER	COACH_NO	TRAIN_ID	STATION_NAME
R001	P001	10-JAN-26	New Delhi	Mumbai Central	AC1	4500	12	A1	T001	Mumbai Central
R002	P001	14-FEB-26	Howrah	New Delhi	AC2	3200	25	B2	T003	New Delhi
R003	P002	20-JAN-26	Chennai Central	Bangalore City	AC3	1200	8	C1	T002	Bangalore City
R005	P003	05-MAR-26	Howrah	New Delhi	AC1	4700	3	A1	T003	New Delhi
R007	P005	18-JAN-26	New Delhi	Mumbai Central	AC2	3100	15	B1	T001	Mumbai Central
R008	P006	20-FEB-26	Howrah	New Delhi	AC3	2100	22	C2	T003	New Delhi
R009	P006	12-MAR-26	Chennai Central	Bangalore City	AC1	1800	5	A1	T002	Bangalore City
R011	P008	08-FEB-26	Howrah	New Delhi	AC2	3300	18	B3	T003	New Delhi
R012	P009	22-JAN-26	Chennai Central	Bangalore City	AC3	1250	9	C1	T002	Bangalore City
R013	P009	01-MAR-26	New Delhi	Mumbai Central	AC1	4600	2	A1	T001	Mumbai Central
R015	P010	08-MAR-26	Chennai Central	Bangalore City	AC2	1900	11	B1	T002	Bangalore City

11 rows returned in 0.00 seconds

CSV Export

3. CONCLUSION

The Railway Reservation Management System provides a structured relational database solution for managing core railway reservation information. The design organizes passengers, passenger phone numbers, trains, stations, reservations, tickets, and payments into related tables with appropriate primary and foreign keys, directly reflecting the entities and relationships in the ER diagram.

The database design supports data integrity, reduces redundancy through normalization up to 3NF, uses derived views for computed attributes such as passenger age and train available-seat count, and provides SQL-based retrieval for common railway reservation tasks. The system can serve as a backend foundation for a railway booking application in which users can search trains, manage passenger details, reserve seats, and track tickets and payments.

4. FUTURE ENHANCEMENTS

- Role-based access for administrators, reservation staff, and other authorized users.
- Cancellation, refund, RAC, and wait-list management.
- Real-time seat availability and automatic seat allocation.
- Online payment gateway integration and automated payment reconciliation.
- SMS/email/app notifications for booking and journey updates.
- Train route and timetable optimization.
- Passenger travel history and loyalty management.
- Automated reports and analytics dashboards.
- Backup, recovery, audit logging, and stronger database security.
- Web and mobile application interfaces.

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